

内容概述

NOKOV SDK 使用手册用于帮助开发者了解 SDK 的使用规则，该手册包含所有的接口及详细说明，为用户最终实现从 NOKOV 程序接收动捕数据。

适用的 NOKOV 程序及版本：

XINGYING 1.0.0.4321 及以上版本；
XING 1.4.0.4537 及以上版本；

一、类型定义

1、sRigidBodyData

刚体信息

```
public struct sRigidBodyData {  
    /// <summary> ///  
    The identifier ///  
    </summary>  
    public Int32 Id;  
    /// <summary> ///  
    The x  
    /// </summary>  
    public float X; ///  
    <summary> ///  
    The y  
    /// </summary>  
    public float Y; ///  
    <summary> ///  
    The z  
    /// </summary>  
    public float Z; ///  
    <summary> ///  
    The qx  
    /// </summary>  
    public float QX; ///  
    <summary> ///  
    The qy  
    /// </summary>  
    public float QY;
```

```
/// <summary> /// The qz  
/// </summary> public  
float QZ; /// <summary> ///  
The qw /// </summary>  
public float QW; ///  
<summary> /// The n  
markers /// </summary>  
public int NMarkers; ///  
<summary> /// The  
markers /// </summary>  
public IntPtr Markers; ///  
<summary> /// The marker  
ids /// </summary> public  
IntPtr MarkerIDs; ///  
<summary> /// The marker  
sizes /// </summary>  
public IntPtr MarkerSizes;  
/// <summary> /// The  
mean error /// </summary>  
public float MeanError; ///  
<summary> /// The  
parameters /// </summary>  
public Int16 Params;
```

}

Members

ID 刚体 ID

x, y, z 刚体位
置

qx, qy, qz, qw
刚体方向

NMarkers	组成刚体的 Marker 点个数
Markers	Marker 点数据
MarkerIDs	Marker ID
MarkerSizes	Marker 大小
MeanError	偏差
params	保留

2、 sSkeletonData

人体信息。

```
public struct sSkeletonData {
    /// <summary>
    /// The identifier
    /// </summary>
    public Int32 Id;
    /// <summary>
    /// The rigid body count
    /// </summary>
    public Int32 RigidBodyCount;
    /// <summary>
    /// The rigid bodies
    /// </summary>
    public IntPtr RigidBodies; // Pointer to sRigidBodyData[RigidBodyCount]
}
```

Members

ID

人体 ID

RigidBodies

刚体/骨骼个数

sRigidBodyData

刚体/骨骼信息

3、sMarker

Marker 信息。

```
public struct sMarker {  
    /// <summary> ///  
    The identifier ///  
    </summary>  
    public Int32 Id; ///  
    <summary> ///  
    The position x ///  
    </summary>  
    public float X; ///  
    <summary> ///  
    The position y ///  
    </summary>  
    public float Y; ///  
    <summary> ///  
    The position z ///  
    </summary>  
    public float Z; ///  
    <summary> ///  
    The size  
    /// </summary> public float  
    Size; /// <summary> /// The  
    parameters(reserved) ///  
    </summary> public Int16  
    Params;  
}
```

Members

ID Marker 点 ID

x, y, z Marker 点位
置

size Marker 大小

params
保留

4、sFrameOfMocapData

动捕数据。

```
public struct sFrameOfMocapData {  
    /// <summary> /// The frame  
    number /// </summary>  
    public Int32 FrameNumber;  
  
    /// <summary>  
    /// The marker set count  
    /// </summary>  
    public Int32 MarkerSetCount;  
    /// <summary>  
    /// The marker sets  
    /// </summary>  
    [MarshalAs(UnmanagedType.ByValArray, SizeConst =  
NokovSDKConstants.MaxModels)] public  
    sMarkerSetData[] MarkerSets;  
  
    /// <summary>  
    /// The undefined marker count,  
    /// </summary>  
    public Int32 OtherMarkerCount;  
    /// <summary>  
    /// The undefined markers, Pointer to float[OtherMarkerCount][3] ///  
    </summary>  
    public IntPtr OtherMarkers;
```

```
/// <summary>
/// The rigid body count
/// </summary>
public Int32 RigidBodyCount;
/// <summary>
/// The rigid bodies
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxRigidBodies)] public
sRigidBodyData[] RigidBodies;
```

```
/// <summary>
/// The skeleton count
/// </summary>
public Int32 SkeletonCount;
/// <summary>
/// The skeletons
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxSkeletons)]
public sSkeletonData[] Skeletons;
```

```
/// <summary>
/// The labeled marker count
/// </summary>
public Int32 LabeledMarkerCount;
/// <summary>
/// The labeled markers
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxLabeledMarkers)]
public sMarker[] LabeledMarkers;
```

```
/// <summary>
/// The analogdata count
/// </summary>
public Int32 AnalogdataCount;
/// <summary>
/// The analogdata
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxAnalogChannels)]
public float[] Analogdata;
```

```
/// <summary>
/// The host defined time delta between capture and send
/// </summary>
public float FLatency;
/// <summary>
/// The SMPTE timecode (if available)
/// </summary>
public UInt32 Timecode;
/// <summary>
/// The timecode subframe
/// </summary>
public UInt32 TimecodeSubframe;
/// <summary>
/// The FrameGroup timestamp, the number of milliseconds since the Epoch
1970-01-01 00:00:00 +0000 (UTC).
/// </summary> public
UInt64 Timestamp; ///
<summary>
/// The host defined parameters
/// </summary>
public Int16 Params; }
```

Members

- FrameNumber
帧号
- MarkerSetsCount
MarkerSet 个数
- MocapData
MarkerSet 信息
- OtherMarkersCount
未命名 Marker 点个数
- OtherMarkers
未命名 Marker 点数据
- RigidBodiesCount
刚体个数
- RigidBodies

刚体信息

SkeletonsCount 人
体/骨骼个数

Skeletons 人体/骨
骼信息

LabeledMarkersCount
命名 Marker 点个
数

LabeledMarkers 命名
Marker 点数据

Analogdatas 模拟数据
通道个数

Analogdata
模拟数据

FLatency
时间差

Timecode SMPTE
时间码

TimecodeSubframe
帧数偏差

TimeStamp 时间戳
(毫秒)

params
保留

6、 Verbosity

日志等级


```
public enum NokovVerbosity
{
    /// <summary>

    /// The verbosity none

    /// </summary>

    Verbosity_None = 0,

    /// <summary>

    /// The verbosity information

    /// </summary>

    Verbosity_Info, /// <summary>

    /// The verbosity warning ///

    </summary>

    Verbosity_Warning, ///

    <summary> /// The verbosity

    error /// </summary>

    Verbosity_Error,
```

```
/// <summary>

/// The verbosity debug

/// </summary>

Verbosity_Debug,

}
```

7、sMarkerSetDescription

MarkerSet 描述信息。

```
public struct sMarkerSetDescription {
    /// <summary> /// The
    MarkerSet name ///
    </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
    public string Name;
    /// <summary>
    /// The number of markers in MarkerSet
    /// </summary>
    public Int32 MarkerCount;
    /// <summary>
    /// The marker names, char**, "array of marker names" ///
    </summary>
    public IntPtr MarkerNames;
}
```

Members

Name MarkerSet 名称

MarkersCount
Marker 点个数

MarkerNames Marker
点名称数组

8、sRigidBodyDescription

刚体描述信息。

```
public struct sRigidBodyDescription
{
    /// <summary>
    /// The RigidBody name
    /// </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
    public string Name;

    /// <summary>
    /// The RigidBody identifier
    /// </summary>
    public Int32 Id;
    /// <summary>
    /// ID of parent Rigid Body (in case hierarchy exists) ///
    </summary>
    public Int32 ParentId;
    /// <summary>
    /// The offset position x relative to parent ///
    </summary>
    public float OffsetX;
    /// <summary>
    /// The offset position y relative to parent ///
    </summary>
    public float OffsetY;
    /// <summary>
    /// The offset position z relative to parent ///
    </summary>
    public float OffsetZ;
    /// <summary>
    /// The Orientation qx relative to parent ///
    </summary>
    public float Qx;//
    /// <summary>
    /// The Orientation qy relative to parent
```

```
    /// </summary>
    public float Qy;
    /// <summary>
    /// The Orientation qz relative to parent ///
    </summary>
    public float Qz;
    /// <summary>
    /// The Orientation qz relative to parent ///
    </summary>
    public float Qw;
}
```

Members

- Name 刚体名称
- ID 刚体 ID
- parentID 父刚体 ID
- offsetx, offsety, offsetz 相对父刚体的 xyz 坐标
- qx, qy, qz, qw 相对父刚体的方向

9、sSkeletonDescription

人体描述信息。

```
public struct sSkeletonDescription
{
    /// <summary>
    /// The name
    /// </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
```

```
public string Name;

/// <summary> /// The
identifier /// </summary>
public Int32 Id; /// <summary>
/// The rigid body count ///
</summary> public Int32
RigidBodyCount;

/// <summary>
/// The rigid bodies
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxSkeletonRigidBodies)]
public sRigidBodyDescription[] RigidBodies;
}
```

Members

Name

人体名称

ID

人体 ID

RigidBodiesCount

组成人体的刚体个数

RigidBodies

刚体数组

10、sDataDescription

数据描述信息。

```
public struct sDataDescription {
    /// <summary> /// The
description type
```

```
    /// </summary> public Int32
    DescriptionType; /// <summary>
    /// The description pointer ///
    </summary> public IntPtr
    Description;
}
```

Members

type 描述类
型

11、sDataDescriptions

动捕数据描述

```
public struct sDataDescriptions {
    /// <summary>
    /// The data description count
    /// </summary>
    public Int32 DataDescriptionCount;

    /// <summary>
    /// The data descriptions
    /// </summary>
    [MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxModels)]
    public sDataDescription[] DataDescriptions;
}
```

Members

sDataDescriptions
描述数目

DataDescriptions
描述数据数组

12、sMarkerSetData

MarkerSet 信息。

```
public struct sMarkerSetData
{
    /// <summary>
    /// The name
    /// </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
    public string Name;

    /// <summary>
    /// The marker count
    /// </summary>
    public Int32 MarkerCount;
    /// <summary>
    /// The markers, Pointer to float[MarkerCount][3]
    /// </summary>
    public IntPtr Markers;
}
```

Members

- Name MarkerSet 名称
- MarkersCount Marker 点个数
- Markers Marker 点数据

二、接口定义

1、Initialize

初始化 SDK，调用其他 SDK 函数的前提。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
```

```
public static extern NokovSDKError Initialize(IntPtr clientHandle, string
szServerAddress);
```

Parameters

szServerAddress [in] 服务器 ip 地址

2、Uninitialize

释放 SDK 资源，在程序结束之前调用。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError Uninitialize(IntPtr clientHandle);
```

3、SetDataCallback

设置动捕数据回调。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError SetDataCallback(IntPtr clientHandle,
NokovSDKFrameReceivedCallback dataCallback, IntPtr pUserData);
```

Parameters dataCallback [in]

动捕数据回调函数

pUserData [in] 用户数据

Callback Function Parameters

pUserData [out] 用户数据

4、SetMessageCallback

设置日志输出回调。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError SetMessageCallback(IntPtr
clientHandle, NokovSDKMessageCallback messageCallBack);
```


);

Parameters messageCallBack
[in] 动捕数据回调函数

5、GetServerDescription

获取动捕服务应用信息。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =  
NokovSDKConstants.NokovSDKLibCallingConvention)]  
public static extern NokovSDKError GetServerDescription(IntPtr  
pServerDescription);
```

Parameters

pServerDescription [out] 动
捕服务应用信息

6、GetDataDescriptions

获取动捕数据描述。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =  
NokovSDKConstants.NokovSDKLibCallingConvention)]  
public static extern NokovSDKError GetDataDescriptions(IntPtr  
clientHandle, out IntPtr pDataDescriptions);
```

Parameters

pDataDescriptions [out]
动捕数据描述

7、DecodeTimecode

解码时间码。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =  
NokovSDKConstants.NokovSDKLibCallingConvention)]  
public static extern NokovSDKError DecodeTimecode(IntPtr clientHandle, uint  
inTimecode, uint inTimecodeSubframe, ref int hour, ref int minute, ref int second, ref int  
frame, ref int subframe);
```

Parameters

inTimecode [in]
时间码

inTimecodeSubframe
[in] 帧数偏差

hour
[out] 时

minute
[out] 分

second
[out] 秒

frame
[out] 帧数 [1-30]

subframe
[out] 帧数偏差

Return Values

TRUE 表示成功，FALSE 表示失败。

8、TimecodeStringify

解码时间码后字符串格式化。.

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError TimecodeStringify(IntPtr clientHandle,
uint inTimecode, uint inTimecodeSubframe, string Buffer, int BufferSize);
```

Parameters

inTimecode
[in] 时间码

inTimecodeSubframe
[in] 帧数偏差

Buffer [out] 缓存

BufferSize [in] 缓存大小

Return Values

TRUE 表示成功，FALSE 表示失败。

9、NokovFreeFrame

动捕数据内存释放。

[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention = NokovSDKConstants.NokovSDKLibCallingConvention)]

public static extern NokovSDKError NokovFreeFrame(IntPtr clientHandle, IntPtr pFrameOfMocapData);

Parameters

pFrameOfMocapData
[in] 动捕数据

10、GetLastFrameOfMocapData

获取一帧动捕数据。

[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention = NokovSDKConstants.NokovSDKLibCallingConvention)]

public static extern IntPtr GetLastFrameOfMocapData(IntPtr clientHandle);

Return Values

返回一帧动捕数据，NULL 失败。

Remarks

返回的数据在使用后需要调用 [NokovFreeFrame](#) 进行内存释放。

非线程安全。